LAB 7 : Configuration of Switch, VLAN configuration and Inter-VLAN Routing

OBJECTIVES :

- To create and configure multiple VLANs on a network switch.
- To configure trunking and assign switch ports to appropriate VLANs.
- To implement and verify Inter-VLAN Routing using a router.

THEORY :

1) **Switch Configuration :**

Virtual Local Area Network (VLAN) :

A switch is a Layer 2 networking device that connects multiple devices within a local area network (LAN). It forwards data frames using MAC addresses, allowing communication to occur only between the intended devices, which improves efficiency. Configuring a switch includes setting up interfaces, managing ports by enabling or disabling them, and assigning ports to appropriate VLANs. Proper configuration minimizes collisions, enhances overall network performance, and ensures better organization and control of network traffic..

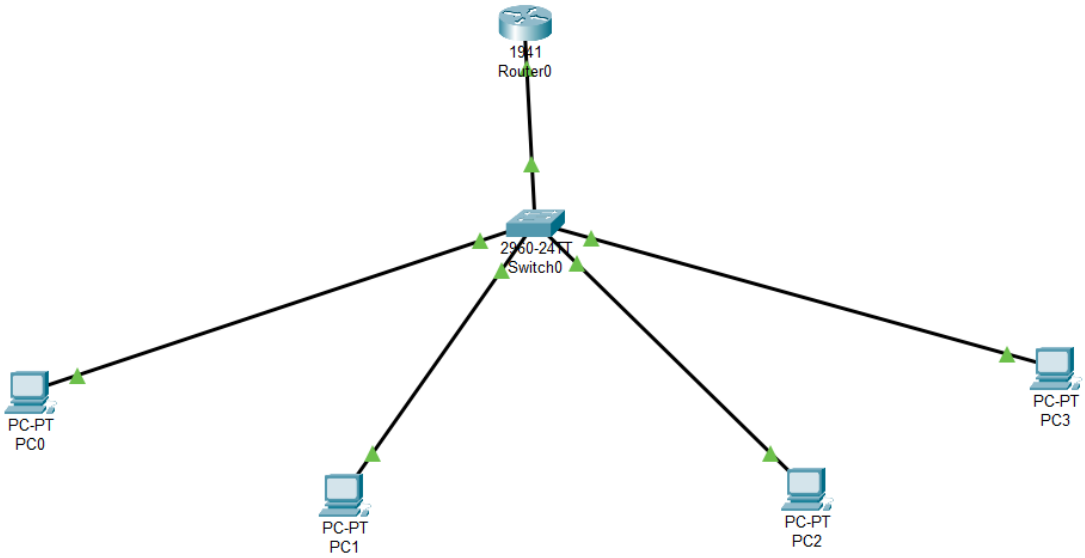
2) **VLAN Configuration :**

A VLAN (Virtual Local Area Network) is a logical method of dividing a network within a switch. It allows devices to be grouped together logically, even if they are connected to different physical ports. This separation limits broadcast traffic, enhances security, and makes network management more organized and flexible. Each VLAN functions as its own separate broadcast domain.

3) **Inter-VLAN Routing :**

Inter-VLAN routing is the process of forwarding network traffic between different VLANs. Since VLANs are separate broadcast domains, a Layer 3 device such as a router is required for communication between them. In this method, a single physical router interface is divided into multiple sub-interfaces, each configured for a specific VLAN. This approach is commonly known as “Router-on-a-Stick.”

Network Topology :



Observation :

```
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=2ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

Fig: Ping in same LAN

```
C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Reply from 192.168.20.3: bytes=32 time<1ms TTL=127
Reply from 192.168.20.3: bytes=32 time<1ms TTL=127
Reply from 192.168.20.3: bytes=32 time<1ms TTL=127
Reply from 192.168.20.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Fig: Ping in Different LAN

Configuration :

VLAN and IP Addressing Scheme :

VLAN	Name	Network	Gateway
10	Computer	192.168.10.0/24	192.168.10.1
20	Electronics	192.168.20.0/24	192.168.20.1

Device	IPv4 address	Subnet Mask	Default Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
PC2	192.168.20.2	255.255.255.0	192.168.20.1
PC3	192.168.20.3	255.255.255.0	192.168.20.1

Commands :

Switch Configuration :

1. **Create VLANS**

   Switch(config)# vlan 10

   Switch(config-vlan)# name Computer

   # exit

   Switch(config)# vlan 20

   Switch(config-vlan)# name Electronics

   # exit

2. **Assign port to VLANS** Switch(config)# interface range fa0/1-2

   # switchport mode access

   # switchport access vlan 10

   # exit

   # interface range fa0/3-4

switchport mode access

switchport access vlan 20

exit

3. **Configure Trunk Port (Switch-Router)**

Switch(config)# interface fa0/5

switchport mode trunk

exit

Verification Switch :

Switch# show vlan brief

```
Switch>enable
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   computer                         active    Fa0/1, Fa0/2
20   electronics                      active    Fa0/3, Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Router Configuration (Inter-VLAN Routing) :

Router(config)# interface GigabitEthernet 0/0.10

encapsulation dot1Q 10

ip address 192.168.10.1 255.255.255.0

exit

Router(config)# interface GigabitEthernet 0/0.20

encapsulation dot1Q 20

ip address 192.168.20.1 255.255.255.0

```
# exit
```

RESULT :

In this experiment, the switch was configured with multiple VLANs, and each port was properly assigned to its designated VLAN. A trunk link was set up between the switch and the router to carry traffic from multiple VLANs. Inter-VLAN routing was achieved by configuring sub-interfaces on the router. As a result, devices within the same VLAN communicated successfully, and communication between different VLANs was enabled through the router.

DISCUSSION :

The VLAN configuration logically divided the network into separate broadcast domains, reducing unnecessary broadcast traffic and enhancing overall efficiency. By assigning switch ports to specific VLANs, devices were organized based on their roles or functions. The trunk port carried traffic from multiple VLANs between the switch and the router through VLAN tagging. Inter-VLAN routing allowed communication between different VLANs in a controlled manner while preserving proper network segmentation. Verification commands were used to ensure that the VLANs, trunk connections, and routing settings were correctly implemented.

CONCLUSION :

The objectives of configuring VLANs on a switch and enabling inter-VLAN routing using a router were successfully implemented and verified.